

Software Design Description 300

ESP32-S3 Receiver – Overview

Overview

The ESP32-S3 is the receiver end of the trail camera system. It receives compressed WHT coefficients from the DevKitV1 over LoRa, reconstructs frames using inverse WHT and Gaussian smoothing against stored reference frames, and serves the reconstructed images to a mobile app via WiFi AP and WebSocket.

Hardware: ESP32-S3 WROOM | 8 MB PSRAM | UART to LoRa (RYLR998) | WiFi AP | No SD card

Module Hierarchy

main.c	Entry point, init, task creation, heap monitor
---- lora_uart.c	UART interface to RYLR998 LoRa module
---- lora_bench.c	LoRa bench test echo slave (TEST_MODE_LORA_BENCH)
---- image_reconstruct.c	Gaussian reconstruction from WHT coefficients
---- wifi_ap.c	WiFi Access Point ("TrailCamera")
\---- ws_server.c	WebSocket server - streams frames to Flutter app

Operating Modes

Production Mode (default)

Two functional domains, pinned to separate cores:

Domain	Core	Tasks	Purpose
LoRa + Reconstruction	0	lora_rx	Receive packets, reconstruct frames
WiFi + WebSocket	1	httpd (async)	Serve frames to mobile clients

LoRa Bench Mode (`#define TEST_MODE_LORA_BENCH`)

Only the `lora_bench` echo slave task runs. WiFi and WebSocket are disabled. The S3 echoes received packets back to the DevKitV1 for RF link validation. Includes ready/start handshake protocol. See SDD004.

LoRa Interface

Parameter	Value
UART	UART_NUM_2
TX pin	GPIO 17 (ESP32 TX → module RX)
RX pin	GPIO 18 (module TX → ESP32 RX)
Baud	115200
Module address	1 (receiver)
Network ID	6 (must match DevKitV1)
Max payload	240 bytes (RYLR998 limit)

Critical wiring note: Swapping TX/RX on the S3 (GPIO17/GPIO18) silently kills all comms with no error output. See SDD004 Bug 1 context.

Image Reconstruction

Reference Frame Storage

- 3 pan/tilt positions stored in PSRAM
- 3 x 307,200 bytes (640x480 grayscale) = ~921 KB
- Allocated with `heap_caps_malloc(MALLOC_CAP_SPIRAM)` at init

Reconstruction Pipeline

Maps to `sim/codec.py:decode_frame()`:

1. **Parse packet header** — extract position index, scale, bounding box
2. **Identify active blocks** — blocks with nonzero WHT coefficients
3. **Dequantize** — multiply quantized values by `q_step`
4. **Inverse 2D WHT** — same butterfly as forward transform, divide by 64
5. **Gaussian smoothing** — smooth reconstructed diff to reduce block artifacts
6. **Add to reference** — `frame_out = reference[position] + smoothed_diff`
7. **Clamp** — pixel values to `[0, 255]`

Frame Constants

Parameter	Value	Notes
FRAME_W	640	Pixels per row
FRAME_H	480	Rows per frame
FRAME_BYTES	307,200	Grayscale
BLOCK_SIZE	8	WHT block dimension
BLOCKS_W	80	Blocks per row
BLOCKS_H	60	Block rows per frame
NUM_POSITIONS	3	Pan/tilt positions

WiFi Access Point

Parameter	Value
SSID	TrailCamera
Password	wildlife1 (WPA2)
Channel	6
Max connections	4
IP	192.168.4.1

- Created on boot using ESP-IDF WiFi AP API
- NVS flash initialized for WiFi state
- Event handler logs client connect/disconnect with MAC address

WebSocket Server

Parameter	Value
Port	80
Path	/stream
Max clients	4

Parameter	Value
Protocol	Binary WebSocket

Frame format (binary message):

Byte offset	Size	Content
0	1	Position index (0=left, 1=center, 2=right)
1-4	4	Frame sequence number (uint32)
5+	FRAME_BYTES	Raw grayscale pixels, row-major

Push-based: `ws_broadcast()` sends to all connected clients when a new frame is reconstructed. Disconnected clients are cleaned up on send failure.

Source Files

File	Purpose	Status
<code>src/main.c</code>	Init, mode selection, task creation	Done
<code>src/lor_a_uart.c</code>	LoRa UART AT command interface	Done
<code>src/lor_a_bench.c</code>	Bench test echo slave + handshake	Done
<code>src/image_reconstruct.c</code>	Gaussian reconstruction from coefficients	Done (needs WHT coefficient parsing update)
<code>src/wifi_ap.c</code>	WiFi AP creation	Done
<code>src/ws_server.c</code>	WebSocket frame broadcast	Done

Known Issues

1. **LoRa bench mode still enabled** — `TEST_MODE_LORA_BENCH` is `#define`'d in `config.h`. Must be commented out to enable production mode (WiFi + reconstruction).
2. **Power rail sag at high TX power** — S3 dev board LDO cannot supply RYLR998 at `CRFOP=22`. Currently limited to `CRFOP=7`. Needs bulk capacitor for field deployment (SDD004, Bug 5).
3. **Frame constants mismatch** — `config.h` defines 640x480 but WHT pipeline operates on 320x240. Needs alignment when FPGA pipeline is integrated.

References

- SDD000 — Development Standards
- SDD003 — LoRa Link Budget
- SDD004 — LoRa Bench Test & Debug
- SDD200 — ESP32-DevKitV1
- SDD500 — Simulation Pipeline
- `sim/codec.py:decode_frame()` — Reference reconstruction implementation